

# Amanuel Ergogo

PhD Student (CSE) — Robotics Core, Robot Learning, Simulation, Teleoperation, ML Control Policies  
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## SUMMARY

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Robotics and Machine Learning researcher with experience in robotics simulation, VR/AR, imitation learning, reinforcement learning, and Sim-to-real. I developed a virtual environment for Isaac Sim extension for granular-material manipulation, including robotic arms/end-effectors, a VR interface for high-quality demonstration collection, hand-tracking-to-robot motion retargeting, domain randomization, and integration with IL/RL training pipelines. I trained and evaluated IL and RL policies for scooping and transport tasks. The results are published at the IROS 2025 [Paper](#) + [video](#).

## TECHNICAL SKILLS

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**Robot Learning:** Imitation Learning, Reinforcement Learning, Vision-Language-Action (VLA), PyTorch, CUDA

**Robotics & Control:** ROS/ROS2, MoveIt, Kinematics, Dynamics, Motion Retargeting, Motion Planning, Task Planning

**Simulation:** Isaac Sim, Isaac Lab, MuJoCo

**Software & Systems:** Python, C/C++, Git, Linux, Docker, SLURM

**VR/HRI:** Unity3D

## EXPERIENCE

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**Research Assistant** Jan 2025 – Present  
Reality, Autonomy, and Robot Experience (RARE) Lab — Tampa, FL

- Developed methods to explain IL/RL policy behavior and failures for safety-critical applications; analyzed predicted actions and error modes.
- Designed imitation learning tasks, collected datasets, and trained/evaluated an **action-chunking Transformer and Diffusion** based imitation learning model.
- Collaborated with a research team to plan experiments and communicate progress/results.

**AI Robotics Researcher** Apr 2022 – Dec 2024  
Sano Center for Computational Personalized Medicine — Poland

- Built a **human-in-the-loop imitation learning** system enabling experts to teach assistive manipulation tasks; implemented training + evaluation loops.
- Developed a **VR teleoperation + data collection pipeline** (Quest 2, HoloLens 2) for robotic manipulation learning with multimodal sensing (RGB-D, force).
- Trained and benchmarked **IL (robomimic)** and **RL** control policies across simulation (IsaacSim/Gazebo) and real robot rollouts.
- Deployed policies to a 7DOF UFactory xArm via **ROS**; conducted sim-to-real transfer testing and evaluation.

## EDUCATION

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**Ph.D., Computer Science and Engineering**, University of South Florida (Advisor: Prof. Zhao Han)  
Jan 2025 – Present

**M.Eng., Control Engineering and Robotics**, Wroclaw University of Science and Technology, Poland  
Feb 2020 – Oct 2021

**B.S., Electrical and Computer Engineering**, Wolaita Sodo University, Ethiopia Sep 2013 – Jul 2018

## SELECTED PUBLICATIONS

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- “VERAGMIL: Virtual Environment for Scooping Granular Foods with Imitation Learning Models.” **Amanuel Ergogo\***, D. Dall’Alba, P. Korzeniowski. *IROS 2025*. [Accepted]
- “HiLoTEL: Virtual Reality Robot Interface-Based Human-in-the-Loop Task Execution and Learning...” **Amanuel Ergogo\***, D. Dall’Alba, P. Korzeniowski. *IEEE VR*, 2024. [Accepted]
- “Human–Robot Collaboration Failures in Generative Visuomotor Models.” **Amanuel Ergogo\***, Grant Henderson, Miguel Osorio Vela, Vidhi Mudaliar, Zhao Han. *RSS 2026*. [Under Review]
- “Towards Embodied Agent Intent Explanation in Human-Robot Collaboration...” **Amanuel Ergogo\***, Z. Han. *ICRA 2025 Workshop (HCRL)*, 2025. [Accepted]

\*First author